

Jasper Sands

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RESEARCH INTERESTS

Quantum error correction and decoding · quantum algorithms and query complexity · machine learning for quantum-device characterization and control.

EDUCATION

Columbia University, New York, NY Sep 2025 – May 2027

Master of Science in Computer Science, Quantum Computing Track. GPA 3.95/4.0.

Thesis: *Automation of Silicon Spin Qubit Measurements Using Machine Learning.* Advised by Dr. Nard Dumoulin Stuyck at Diraq and UNSW Sydney, with Dr. Henry Yuen at Columbia.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & Machine Learning · Neural Networks & Deep Learning · Machine Learning for Functional Genomics.

Washington University in St. Louis, St. Louis, MO Aug 2021 – May 2025

Bachelor of Science in Computer Science and Economics; Minor in Political Science. GPA 3.81/4.0.

Coursework: Machine Learning · Analysis of Algorithms · Algorithms for Biosequence Comparison.

TECHNICAL REPORTS

Mike Azure and Jasper Sands. *Practical LLM-Based Lossless Compression: An Empirical Study.* Columbia University, 2025.

John C. Papciak and Jasper Sands. *Multimodal Hyperbolic Embeddings for AMR Detection and Taxonomic Modeling in Metagenomic Contigs.* Columbia University, 2025.

RESEARCH EXPERIENCE

Diraq and UNSW Sydney · *MS Thesis Research* Sep 2026 – May 2027

- Automating silicon spin-qubit characterization under Dr. Nard Dumoulin Stuyck, targeting manual tuning and interpretation of charge measurements that bottleneck semiconductor quantum-hardware development.
- Building pipelines that acquire charge-measurement data from cryogenic probing infrastructure and classify charge transitions and device states with machine learning.
- Benchmarking automated classification against expert manual interpretation of the same device measurements.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and optimizing precision under a fixed false-positive budget.

Desdr Open Insurance Toolkit, Columbia University · *Graduate Researcher* Sep 2025 – Dec 2025

- Built climate-risk and actuarial features from sparse SMS survey data with Dr. Eugene Wu, modeling expected agricultural losses for micro-insurance program design.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* May 2025 – Dec 2025

- Reduced hallucination in generative protein modeling with Dr. Venkat Venkatasubramanian by grounding model outputs in ontology and molecular-graph data.

Foster System Database, Washington University in St. Louis · *Undergraduate Researcher* Sep 2024 – May 2025

- Fine-tuned LLaMA-3 under Dr. Ian Fillmore for question answering over 50k foster-care policy documents, with human-in-the-loop feedback for cross-state policy analysis.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for Value at Risk · *CUDA-Q, C++* MIT iQuHACK 2026

- Wrote GPU-accelerated state preparation, threshold oracle, and bisection search for Iterative Quantum Amplitude Estimation; measured query scaling matched the $O(1/\epsilon)$ bound against an $O(1/\epsilon^2)$ classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly* GitHub

- Built a browser-native symplectic stabilizer simulator for rotated and XZZX surface codes with real-time syndrome visualization and MWPM/Union-Find decoding; finite-size scaling estimated thresholds of 3.2% and 2.7% respectively.

Q-Search: multi-agent quantum algorithm research and verification engine GitHub

- Built a multi-agent quantum-algorithm research engine that proposes candidate algorithms, verifies circuits through simulation, benchmarks query and gate complexity against 1,200+ classical baselines, and indexes falsified approaches across hidden-subgroup, dihedral-coset, and code-equivalence problems.

Noise-Aware Adiabatic Simulation: transverse-field Ising model · *Qiskit* IBM Heron hardware

- Digitized adiabatic ground-state preparation on IBM Heron's native fractional RZZ gates, roughly halving logical-to-physical translation error, with palindromic Trotter steps for second-order accuracy.
- Flattened the annealing schedule near the critical point to slow evolution as the spectral gap closes.

TEACHING EXPERIENCE

Introduction to Quantum Computing, Columbia University · *Teaching Assistant* Jan 2026 – May 2026

Parallel and Concurrent Programming, Washington University · *Teaching Assistant* Jan 2025 – May 2025

Reverse Engineering and Malware Analysis, Washington University · *Teaching Assistant* Aug 2024 – Dec 2024

INDUSTRY EXPERIENCE

Smack Technologies · *AI Intern* May 2026 – Aug 2026

- Built physics and sensor simulations for core AI pipelines, integrating knowledge-graph state representations to expand synthetic corpus generation.
- Benchmarked open-weight models across task performance, inference speed, and concurrency to increase generation throughput 64×, and validated simulator outputs against independent references within 0.55% error.

Highnote · *Cybersecurity Engineer Intern* May 2024 – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, ingesting 1+ TB/day and supporting 100+ rule-based and ML anomaly detections.

Mindtrip · *Software Engineer Intern* May 2023 – Jul 2023

- Built internal admin and secure-query tooling with Retool, JavaScript, and Ruby on Rails, saving roughly 250 engineering hours/month.

AWARDS AND COMPETITIONS

- **Overall Winner & NVIDIA Ecosystem Award**, MIT iQuHACK 2026, State Street/Classiq Challenge.
- **Qualified**, NYU Abu Dhabi Quantum Hackathon for Social Good 2027.
- **Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026.
- **Runner-Up**, Qualcomm Snapdragon Multiverse Hackathon 2026.

TECHNICAL SKILLS

Quantum software: Qiskit, Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP · statevector and noise-aware simulation, transpilation, backend execution, benchmarking.

Quantum algorithms: Hamiltonian simulation, quantum phase estimation, amplitude amplification and estimation, QSVT, block encoding, quantum walks, VQE, QAOA, Trotter decomposition.

Quantum error correction: Rotated and XZZX surface codes, stabilizer formalism, MWPM and Union-Find decoding, finite-size scaling, threshold estimation, noise modeling.

Programming: Python, C++, Rust, C, MATLAB, JavaScript/TypeScript, Go, Java.

ML & systems: PyTorch, TensorFlow, scikit-learn, Hugging Face, CUDA, NumPy, SciPy, Linux, Git, Docker, WebAssembly, AWS, GCP, LaTeX.

LEADERSHIP AND SERVICE

- **Graduate Lead**, Columbia Quantum Algorithms Reading Group, Sep 2025 – Jun 2026. Led weekly sessions covering all 33 chapters of Andrew Childs' quantum algorithms notes, paired with simulation checks.
- **Founding-semester member**, Columbia Autonomous Racing Club. Coordinate sponsor outreach and track testing.
- **Volunteer Tutor**, St. Louis Tutor Me Program. K-12 mathematics and literacy in underserved districts.